Supplementary Material

A Geometric Criterion for Sway and Rocking in Inclined Pile Groups, with Field Identification of an Offshore Wind Turbine Foundation

Nguyen Hoang Bui, Thi Cam Nhung Nguyen, Huu-Quyet Nguyen, Chung Truong Thanh, Hung Nguyen-Dang, José C. Matos

This document collects the material that supports the main paper but is not needed to follow its argument: the quality of the raw records, the simulation that sets the threshold of the first check, the complete list of identified poles, the test of the processing settings, the search for a tower mode, and the setup-by-setup comparison of the two campaigns. Section and table numbers without the prefix S refer to the main paper.

S1 Quality of the raw records

Table S1 gives the statistics of the raw records of the seven setups, and Figure S1 shows a typical record of campaign 2. At one moment a single channel jumps far above its normal level while the others stay quiet. The event rings briefly at a high frequency, as a sensor mounting does when it is struck, and is most likely a knock on or near that sensor rather than a motion of the cap, which would appear on every channel. Most of the signal energy lies in the mains harmonics between 50 and 200 Hz, and less than 1 % in the structural band below 5 Hz. The records of campaign 1 are less steady still: the signal level of some channels changes strongly during a record, and some channels are much weaker than others. Only the three-channel setup of campaign 1 is close to steady broadband noise.

Table S1: Measurement setups and raw-record statistics. *Ch.* is the number of channels and *Min.* the record length in minutes. *Peak*/σ is the largest absolute sample of any channel divided by that channel's standard deviation; *Kurt.* the largest channel kurtosis; $N_{>8\sigma}$ the largest number of samples of one channel beyond 8σ; *Drift* the ratio of largest to smallest channel RMS over ten equal segments; *Spread* the ratio of the largest to the smallest channel standard deviation within a setup. The largest value of each column is in bold

Camp.	Setup	Ch.	Samples	Min.	Peak/σ	Kurt.	$N_{>8\sigma}$	Drift	Spread
1	Vertical	5	2 535 936	25.6	133	562	2238	3.2	6.1
1	Lateral 1	5	1 523 873	15.4	25	54	**3131**	12.9	**47.0**
1	Lateral 2	5	1 385 189	14.0	50	1812	1063	**112**	9.2
1	Three-channel	3	1 545 336	15.6	10	4	9	1.5	33.7
2	Lateral	6	906 399	9.1	**139**	**3883**	724	4.7	11.5
2	Vertical d1	6	1 875 536	18.9	69	95	695	4.4	4.1
2	Vertical d2ke	6	2 260 219	22.8	67	88	2564	4.3	4.2

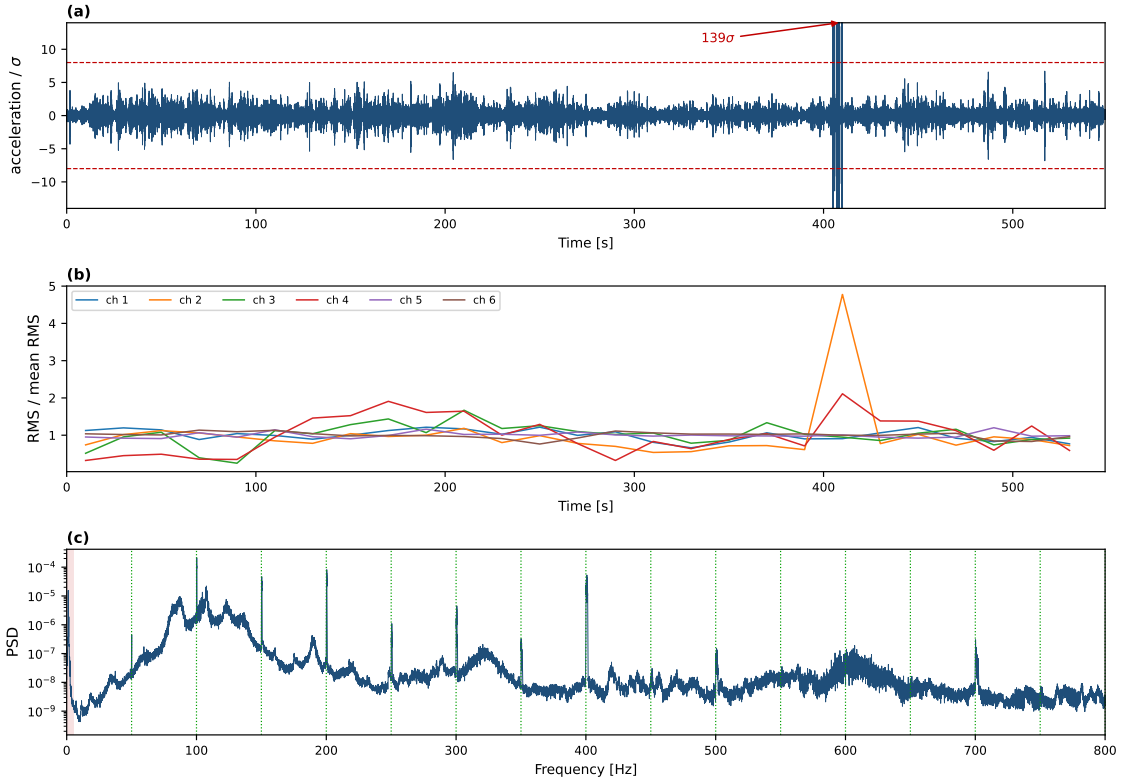


Figure S1: One record of campaign 2. (a) Time history of the channel with the largest event (arrow), in units of its standard deviation. (b) Running RMS of every channel; the event appears on one channel only. (c) Spectrum averaged over the channels, with multiples of 50 Hz dotted and the structural band shaded

S2 Threshold of the effective channel count

The first check of Section 4.2 rejects a pole whose effective channel count $N_{\mathrm{eff}} = (\sum |\phi|^2)^2 / \sum |\phi|^4$ is below 1.25. The threshold was set by simulation. Two cases were generated with the sampling rate, the record length of the shortest record (28 000 samples at 51.61 Hz) and the spectral resolution of the measurements:

- *local response*: a narrow-band signal at 1.2 Hz with 2 % damping on one channel only, and independent white noise on every channel;
- *mode of the cap*: the same signal shared by all five or six channels with the measured shape of the low band, in which opposite corners move in antiphase and the centre hardly moves, again with independent noise on every channel.

Each case was repeated 400 times, at signal-to-noise ratios of 3 and 10, and N_{eff} was computed exactly as for the measurements, from the first singular vector of the cross-power spectral density matrix at the peak of the 1.0–1.4 Hz band. A signal on a single channel never gave more than 1.00. A mode shared by the channels gave a median of 1.55 with five channels and 1.77 with six, and 1.5 or more in 95 % of the trials, at either signal-to-noise ratio. The threshold of 1.25 lies between the two cases.

Two sensitivities were also examined. A difference of gain of up to 3 % between channels, the largest difference between the sensitivities listed in the processing log, changes N_{eff} of the shared mode by at most 4 % (1.49 to 1.62), which leaves it well above the threshold. Noise levels that differ between channels by up to a factor of ten are more harmful: the count of a real mode then fell below 1.25 in 21 % of the trials, so the check can wrongly reject a real mode in such a

record. In the same trials a local response never exceeded 1.01, so the check did not accept one. With three channels the two cases cannot be told apart, and the threshold is not applied to the three-channel setup.

S3 Complete list of identified poles

Table S2 lists every pole returned below 25 Hz, by setup. Table S3 gives the phase between opposite corners of the instrumented frame at the peak of the low band, which underlies Figure 7 of the main paper.

Table S2: Poles returned by the identification, by setup; frequency in Hz with damping ratio in per cent in parentheses

Camp.	Setup	Low band	Higher band
1	Vertical	1.174 (4.74)	18.132 (0.70), 18.565 (0.68), 19.077 (0.58), 19.466 (0.59), 19.920 (1.30)
1	Lateral 1	1.226 (3.43), 1.254 (2.77), 1.731 (5.49)	21.920 (0.40), 22.751 (0.51), 23.754 (0.30)
1	Lateral 2	1.322 (2.71)	21.574 (5.95), 22.813 (1.25), 23.173 (0.59)
1	Three-channel	1.238 (3.01), 1.331 (4.39), 1.726 (5.35)	14.438 (1.42), 14.996 (0.97)
2	Lateral	1.052 (2.69), 1.215 (1.93), 1.612 (2.42)	none accepted
2	Vertical d1	1.147 (9.85)	11.806 (11.52), 13.871 (3.02), 16.833 (2.73), 17.087 (0.59), 17.673 (1.96), 18.954 (1.08), 19.473 (1.37), 19.689 (1.11), 20.549 (0.54)
2	Vertical d2ke	1.074 (1.25), 1.199 (2.35), 1.231 (2.48)	2.296 (1.19)

Table S3: Phase between opposite corners of the instrumented frame at the low-band peak, and amplitude at the centre relative to the mean of the corners. In the lateral setup, channel 4 is the only vertically oriented channel, which is why the 2–4 pair behaves unlike the 1–3 pair there

Setup	Direction	Peak (Hz)	1 vs 3	2 vs 4	Centre / corners
Vertical, campaign 1	vertical	1.134	−179.1°	−173.3°	0.048 / 0.466
Vertical d1, campaign 2	vertical	1.210	+176.3°	+176.6°	0.424 / 0.517
Vertical d2ke, campaign 2	vertical	1.229	−142.7°	−175.2°	0.183 / 0.518
Lateral, campaign 2	horizontal	1.203	−4.9°	−143.3°	0.927 / 0.757

Poles outside the two main groups. A few poles belong neither to the 1.05–1.33 Hz band nor to the 14–24 Hz set. They fail the second check and are not interpreted: a pole at 1.61–1.73 Hz in three of the seven setups, single poles at 0.69 and 0.83 Hz in one lateral setup each, and a pole at 2.30 Hz in one vertical setup.

Two spectral peaks in campaign 2. The spectral peaks of the two vertical setups of campaign 2 differ in shape (modal assurance criterion 0.09), although the modal assurance

criterion between their corresponding poles is 0.967. The band contains more than one mode, and each setup is dominated by a different member of the group.

S4 Processing settings and the mains harmonics

Subspace identification builds its model from the covariances of the records over a time window $\tau = i\,\Delta t$, set by the number of block rows i. At the acquisition rate of 1651.6 Hz with $i = 150$, the window is only 0.09 s, a tenth of a period at 1 Hz. The effect of this window and of the high-pass cut-off was tested on one record (Table S4 and Figure S2). With the short covariance window of the full rate, no pole is found below 5 Hz, whether or not the high-pass filter is applied (configurations A and C). The 1 Hz band appears only with both the longer window and the low cut-off (configuration B), which is the configuration used in the main paper. Configuration E, which keeps the lower sampling rate but shortens the window to 0.097 s, returns no pole at all, so the window, not the sampling rate, is what matters.

Table S4: Comparison of processing configurations, one record (lateral setup, campaign 2). Configurations A–D form a factorial in window and high-pass cut-off, in which the window changes together with the sampling rate; E differs from B in the window alone

Config.	Sampling rate	High-pass	i	τ	Poles returned
A	1651.6 Hz	5 Hz	150	0.091 s	20, from 57.6 to 700.9 Hz; none below 5 Hz
C	1651.6 Hz	0.3 Hz	150	0.091 s	19, from 80.9 to 604.6 Hz; none below 5 Hz
D	51.6 Hz	5 Hz	120	2.33 s	6, from 6.8 to 20.6 Hz; none below 5 Hz
B	51.6 Hz	0.3 Hz	120	2.33 s	3: 1.052, 1.215, 1.612 Hz
E	51.6 Hz	0.3 Hz	5	0.097 s	none

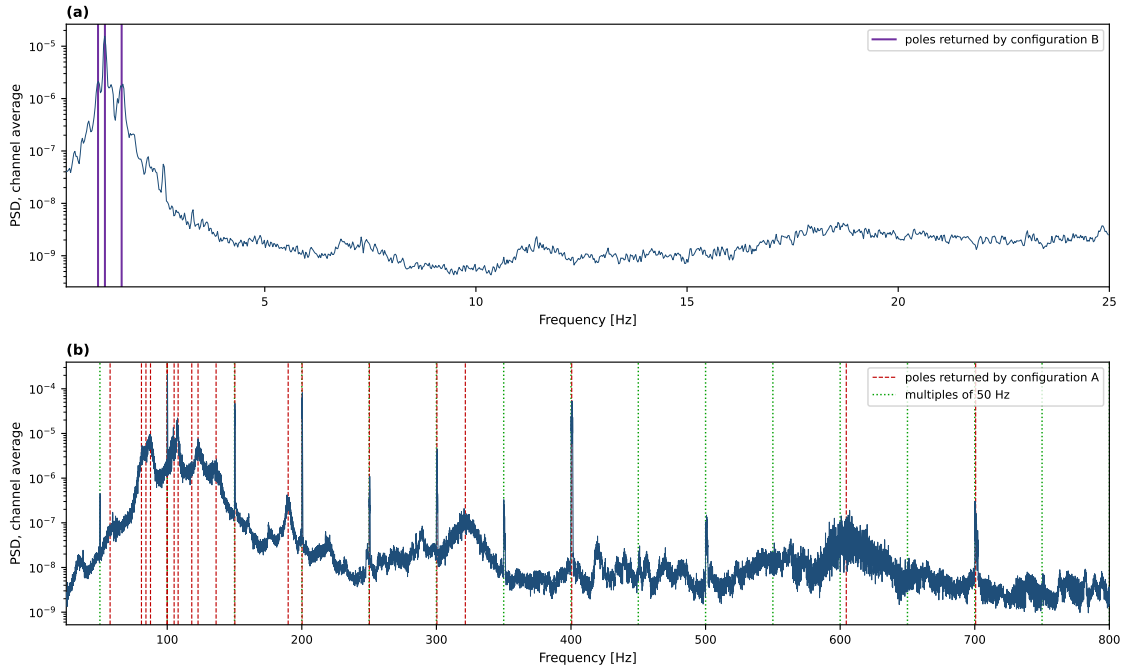


Figure S2: The same record seen by the two processing configurations of Table S4, over (a) the structural band from 0.3 to 25 Hz and (b) the band from 25 to 800 Hz from which configuration A draws its poles. Dotted lines mark multiples of 50 Hz

Seven of the poles returned at the full rate have damping between 0.007 and 0.042 %, far

below the 0.1 % bound of the third check. All seven lie within 0.2 % of a multiple of 50 Hz, at 100.17, 150.30, 200.28, 250.26, 300.49, 400.69 and 700.93 Hz. They are harmonics of the mains supply.

S5 Search for a tower mode below 0.5 Hz

The band below 0.5 Hz was searched for a tower mode, which a soft–stiff design places near 0.3 Hz [1]. The records were decimated further to 12.9 Hz and analysed with a covariance window of 11.6 s (Figure S3). No pole below 0.5 Hz passes the checks. The only features are spectral peaks near 0.21 Hz in campaign 1, recorded before the tower stood and consistent with waves, and near 0.32 Hz in campaign 2, where the first tower mode is expected.

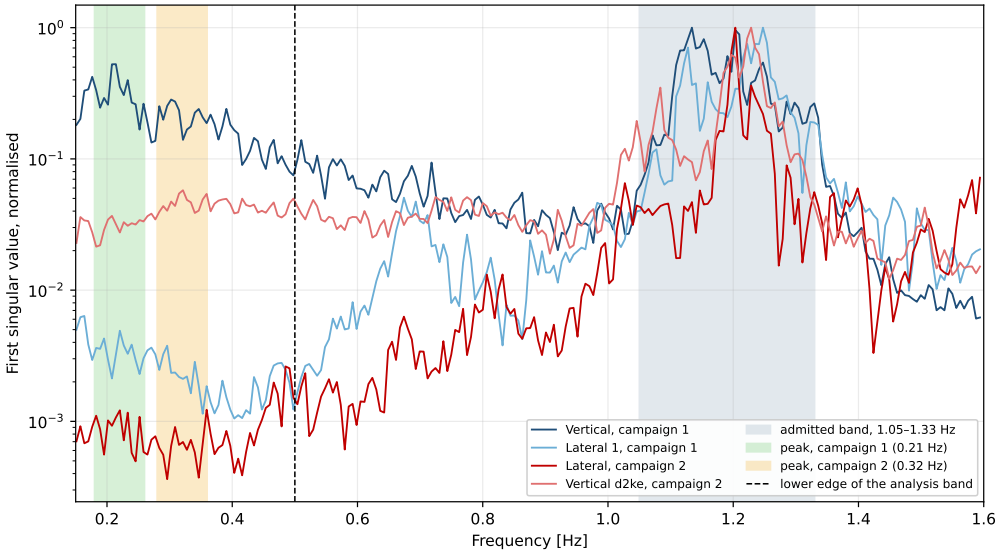


Figure S3: Search below the analysis band, with the records decimated to 12.9 Hz and a covariance window of 11.6 s. The only features below 0.5 Hz are the peaks marked at 0.21 Hz in campaign 1 and 0.32 Hz in campaign 2. The first, recorded before the tower stood, is consistent with wave excitation; the second lies where the first tower mode is expected. Neither yields a stable pole

S6 Setup-by-setup comparison of the two campaigns

Table S5 lists the low-band poles behind Figure 9 of the main paper. Averaging all setups of each campaign suggests a fall of 8.4 % after the tower was installed, but this mixes setups that measure in different directions and therefore weight different members of the band. The table compares setups that measure in the same direction.

Table S5: Low-band poles by setup, grouped by measured direction, and the change between campaigns for like-for-like setups. V: all channels vertical; H: horizontal, with channel 4 vertical in campaign 2; M: one vertical and two horizontal channels

Dir.	Camp.	Setup	Poles (Hz)	Setup mean (Hz)
V	1	Vertical	1.174	1.174
V	2	Vertical d1	1.147	1.147
V	2	Vertical d2ke	1.074, 1.199, 1.231	1.168
		change, vertical:	$-2.3\,\%$ and $-0.5\,\%$	
H	1	Lateral 1	1.226, 1.254	1.240
H	1	Lateral 2	1.322	1.322
H	2	Lateral	1.052 (torsional), 1.215 (translational)	1.134
		change, translational member against lateral 1:	$-2.0\,\%$	
M	1	Three-channel	1.238, 1.331	1.284

References

[1] Arany L, Bhattacharya S, Macdonald J, Hogan SJ (2017) Design of monopiles for offshore wind turbines in 10 steps. Soil Dynamics and Earthquake Engineering 92:126–152. `https://doi.org/10.1016/j.soildyn.2016.09.024`